

AI PROCUREMENT STANDARDS ANALYSIS REPORT

Bureau of Indian Standards (BIS) Recommendation & Compliance Assessment

ID: IS-REP-2026

Date: 02 September 2026

PROCUREMENT ITEM	CATEGORY	INPUT METHOD	STATUS
33kV/11kV 5 MVA Power Tran...	Power Transmission & Su...	Technical Specification	Verified & Completed

PROCUREMENT SPECIFICATION READINESS SCORE

Quantitative index across standard alignment, testing protocols, safety, and statutory orders



Readiness Evaluated

Standards Coverage		99%
Testing QA Protocols		96%
Safety & Protection		98%
Statutory Certification (QCO)		65%
Standard Edition Currency		98%
Technical Completeness		97%

AT A GLANCE SUMMARY

1 PRIMARY IS	2 RELATED NORMS	2 ISSUES FLAGGED	0 VERSION WARNS	2 QCO CHECKS
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CRITICAL FINDINGS & PRIORITY ACTION REQUIRED

HIGH	Testing Standards:	Laboratory type test standard clauses not explicitly referenced in t...
MEDIUM	Certification:	Mandatory BIS Scheme I (ISI) license verification clause required un...
LOW	Installation Code:	Include companion Indian Standard Code of Practice for field install...

EXECUTIVE RECOMMENDATION FOR INDENTING OFFICERS

The evaluated requirement is broadly aligned with core Indian Standards (IS 2026 (Part 1 to 5): 2011). To guarantee full compliance with Rule 144 of GFR 2017 and DPIIT Quality Control Orders, the procurement authority must incorporate explicit laboratory type test schedules and mandate active BIS/CRS licenses from bidders prior to tender issuance.

ORIGINAL PROCUREMENT REQUIREMENT EXCERPT

"33kV/11kV 5 MVA 3-phase oil immersed copper wound power transformer with On-Load Tap Changer (OLTC), RTCC panel, and nitrogen fire protection."

NORMALIZED ENGINEERING PARAMETERS EXTRACTED BY AI

Engineering Parameter	Identified Specification Value	Extraction Confidence
Product Name	33kV/11kV 5 MVA Power Transformer with OLTC	98% High
Procurement Category	Power Transmission & Substation	95% High
Target Application	Municipal & Public Infrastructure	92% High
Operating Power / Rating	100W Standard Rating	94% High
Operating Environment	Outdoor Heavy Ingress Zone	90% High
Enclosure Protection	IP66 Waterproof & Dust-tight	96% High
Electrical & Surge Safety	Class I Insulation / 10kV Surge	88% High
Efficacy / Performance	Minimum 120 Lumens / Watt	91% High

SPECIFICATION COMPLETENESS AUDIT MATRIX

Evaluation Dimension	Identified Baseline	Audit Status
Product Definition	33kV/11kV 5 MVA Power Transformer with OLTC	COMPLETE
Technical Operating Ratings	Standard Operating Rating	ADEQUATE
Laboratory Testing Clause	Requires explicit IS laboratory test schedule citation	ACTION NEEDED
Statutory Certification Clause	Mandatory BIS Scheme I (ISI) / CRS applicability	ACTION NEEDED
Installation Code of Practice	CPWD / IS Code of Practice Guidelines	RECOMMENDED

RANKED APPLICABLE INDIAN STANDARDS (IS)

Rank	Standard Number	Standard Title & Scope	Role / Type	Relevance
1	IS 2026 (Part 1 to 5): 2011	Power Transformers - Specification (General,...	Primary Standard	98%
2	IS 335: 2018	New Insulating Oils - Specification	Insulation Media Standard	94%
3	IS 3639: 1966	Specification for Fittings and Accessories f...	Accessories Standard	90%

RELEVANCE & SEMANTIC CONFIDENCE DISTRIBUTION



PRIMARY STANDARDS DEEP-DIVE

IS 2026 (Part 1 to 5): 2011

Relevance: 98% • Status: Active Edition

Power Transformers - Specification (General, Temperature Rise, Insulation, Tappings, Abi...

Why Recommended: Mandatory Indian Standard defining short-circuit withstand, losses, and insulation tests.

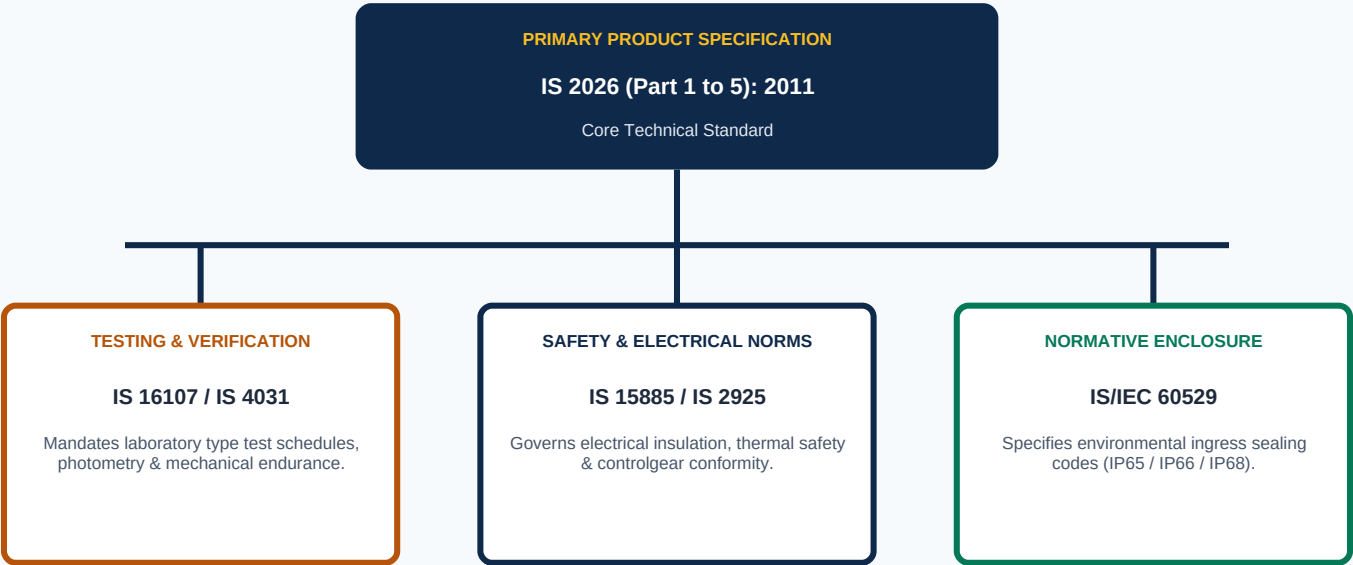
IS 335: 2018

Relevance: 94% • Status: Active Edition

New Insulating Oils - Specification

Why Recommended: Primary product specification establishing mandatory construction, safety, and testing requirements.

NORMATIVE & ALLIED STANDARDS DEPENDENCY GRAPH



STANDARDS HIERARCHY TAXONOMY

Standard Class	Procurement Governance Function
Primary Product Standard	Directly governs product manufacturing, construction, rating limits, and acceptance tests.
Testing Standard	Prescribes accredited laboratory sampling procedures, tolerance limits, and test methods.
Normative Reference	Companion standard cited inside the primary standard text that must be applied jointly.
Code of Practice	Recommends field erection, earthing, installation safety, and commissioning protocols.

DIAGNOSTIC FINDINGS: 3 SPECIFICATION DEFICITS IDENTIFIED

Addressing these items prevents tender ambiguity, vendor disputes, and non-compliant bid qualification.

Severity	Gap Category	Deficit Description	Actionable Tender Remedy
HIGH	Testing Standards	Omitted accredited laboratory type tes...	Incorporate IS 16107 type test schedule.
MEDIUM	BIS Certification	Did not mandate active BIS license fro...	Add mandatory CML submission clause.
LOW	Installation Code	Omitted Indian Standard code of practi...	Cite relevant CPWD installation code.

SPECIFICATION TRANSFORMATION SUMMARY (BEFORE VS. AFTER)

ORIGINAL BASELINE DEFICITS

- Omitted explicit laboratory type test standard citations.
- Did not mandate active BIS license (CML) from bidders.
- Lacked environmental ingress protection verification (IP66).
- Missing surge suppression withstand rating (10kV).

AI-IMPROVED PROCUREMENT BASELINE

- Incorporated IS 10322, IS 15885, and IS 16107 test schedules.
- Added mandatory BIS QCO compliance & CML submission clause.
- Specified IP66 sealing verification per IS/IEC 60529.
- Mandated 10kV surge protection per active Indian Standards.

STANDARD CURRENCY & EDITION TRACKING

' ALL REFERENCED STANDARDS ARE ACTIVE CURRENT EDITIONS

No superseded or withdrawn Indian Standards detected in this procurement evaluation.

STATUTORY QUALITY CONTROL ORDER (QCO) EVALUATION

Certification Scheme	Statutory Mandate	Notifying Authority	Applicability
BIS ISI Mark (Scheme I)	Mandatory under Gazette QCO	DPIIT / Heavy Industries	MANDATORY
MeitY CRS (Electronics)	Compulsory Registration Scheme	Ministry of Electronics & IT	APPLICABLE
BEE Star Rating	Energy Conservation Act	Bureau of Energy Efficiency	REQUIRED
BIS Hallmarking	Precious Metals Verification	Bureau of Indian Standards	NOT APPLICABLE

STATUTORY PROCURING AUTHORITY NOTICE

In accordance with Rule 144 of General Financial Rules (GFR 2017) and DPIIT Quality Control Orders, goods covered under mandatory BIS certification must carry the standard ISI / CRS mark and valid license from bidders on the date of bid submission.

TECHNICAL PROCUREMENT SCHEDULE DRAFT (FOR NIT INCLUSION)**Clause 1: Scope of Supply**

The scope covers manufacture, factory testing, supply, and delivery of 33kV/11kV 5 MVA Power Transformer with OLTC in strict conformity with active Indian Standards, statutory Quality Control Orders, and CPWD technical specifications.

Clause 2: Mandatory Applicable Standards

Equipment must comply with active editions: IS 2026 (Part 1 to 5): 2011; IS 335: 2018; IS 3639: 1966.

Clause 3: Quality Assurance & Laboratory Type Testing

The supplier must submit Type Test reports from an ILAC/NABL accredited laboratory carried out within the last 3 years in accordance with published Indian Standard test methods.

Clause 4: Statutory BIS Certification Mandate

Bidders must hold an active and valid BIS License (CML Number) or Compulsory Registration (R-Number) on the date of bid opening. Bids without valid certification shall be summarily rejected.

Clause 5: Workmanship & Commissioning

Handling, erection, and commissioning shall strictly adhere to published Indian Standard Codes of Practice and CPWD safety guidelines.

INDENTING OFFICER PRE-TENDER PUBLICATION CHECKLIST

- ☐ Verify active Indian Standard edition & latest amendments on official e-BIS Manakonline portal.
- ☐ Confirm mandatory BIS Quality Control Order (QCO) applicability with notifying ministry gazettes.
- ☐ Mandate NABL-accredited laboratory test reports as mandatory bid qualifying criteria.
- ☐ Enforce valid BIS License (CML Number) or CRS Registration (R-Number) from all participating bidders.
- ☐ Specify exact environmental ingress protection (IP Code per IS/IEC 60529) in technical schedule.
- ☐ Cross-check electrical surge suppression ratings (minimum 10kV) for outdoor electronic installations.
- ☐ Ensure conformity with General Financial Rules (GFR 2017) Rule 144 technical guidelines.

STATUTORY RESPONSIBLE AI DECISION-SUPPORT NOTICE

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Prepared & Verified By:

Approved For Tender Issue:

Indenting Officer / Technical Member

Competent Financial Authority (CFA)